Burden of Disease Study (1990 – 2019)

Date: 13th April 2022
FA 550 - Spring 2022

Introduction: Mortality does not give a complete picture of the burden of disease borne by individuals in different populations. The overall burden of disease is assessed using the disability-adjusted life year (DALY), a time-based measure that combines years of life lost due to premature mortality (YLLs) and years of life lost due to time lived in states of less than full health, or years of healthy life lost due to disability (YLDs). This study breaks DALY into its various components to study its trend across time and country.

Research Questions:

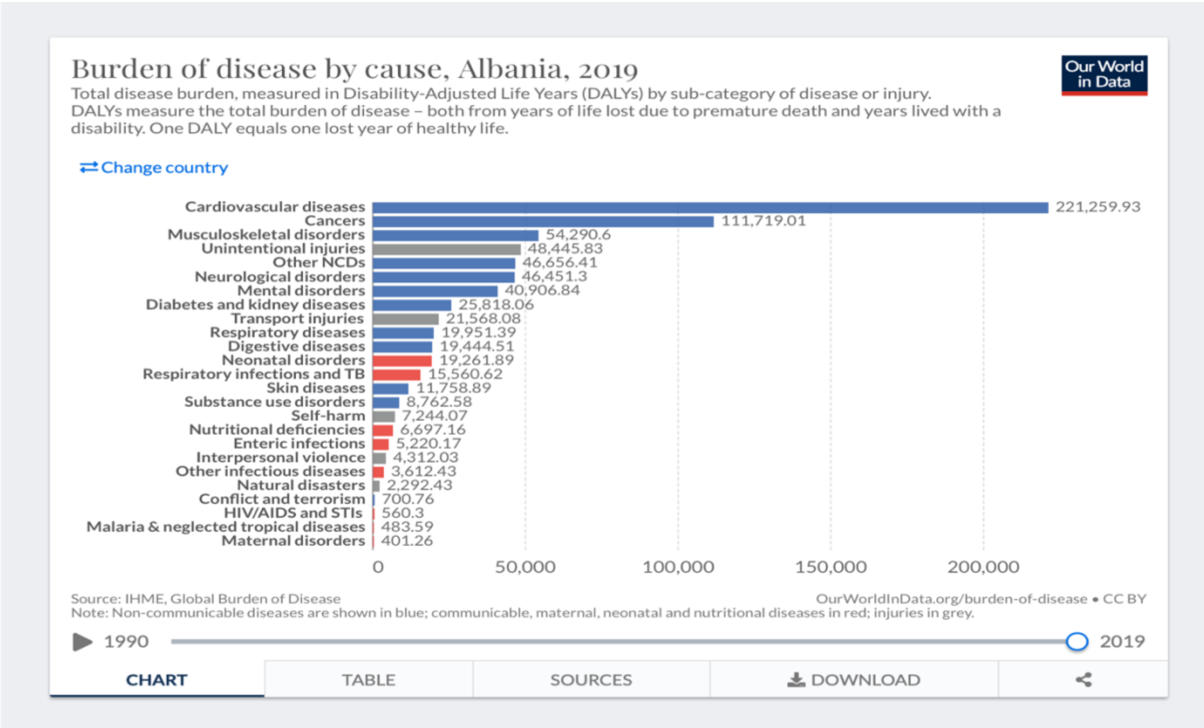
1. Has the burden of disease per country increased or decreased with time?
2. What disease are the major contributing factors of Burden of Disease from 1990 - 2019?
3. Breakdown the top 5 burden of disease by age, non - communicable diseases, communicable, neonatal, maternal and nutritional diseases. Are there any noticeable trends?
4. Report the burden from injuries, violence, self-harm and accidents across countries and time.

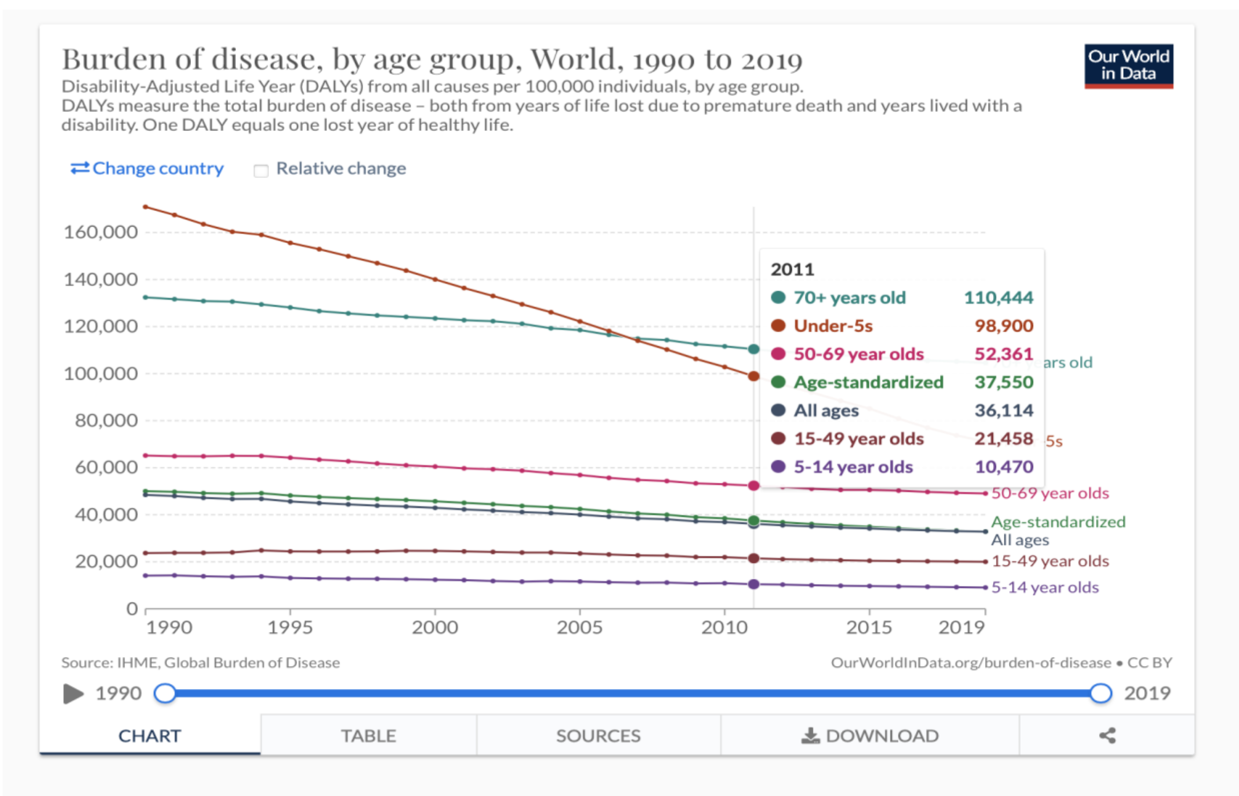
Visualization Tools:

1. Tableau – Tableau provides a host of map and bar graphs. Also, it provides an ease in creating interactive dashboards. Also due to its graphical interface it makes it easy to implement graph designs without the use of extensive coding.

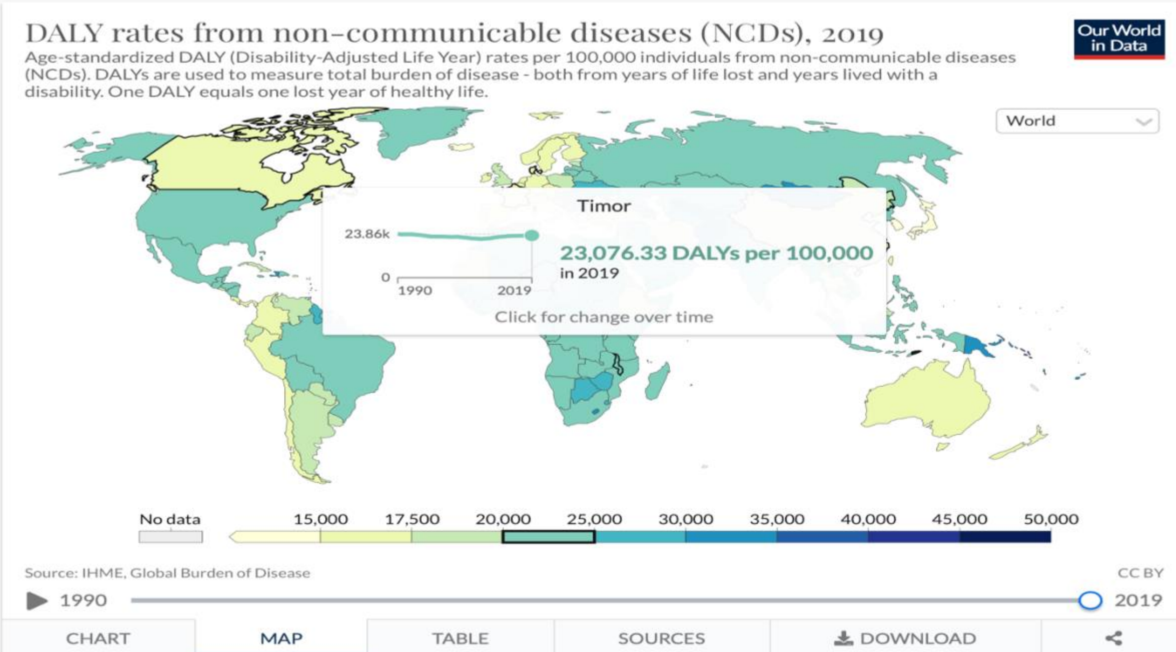
2. R Software – This project will use the gganimate of the ggplot2 package to visualize the time series analysis of the data. R provides a platform to easily shape data in the required format. The html formatting in R can easily merge visualizations from Tableau to create a scrolling story board.

Below are examples of graph chart choices I may use;

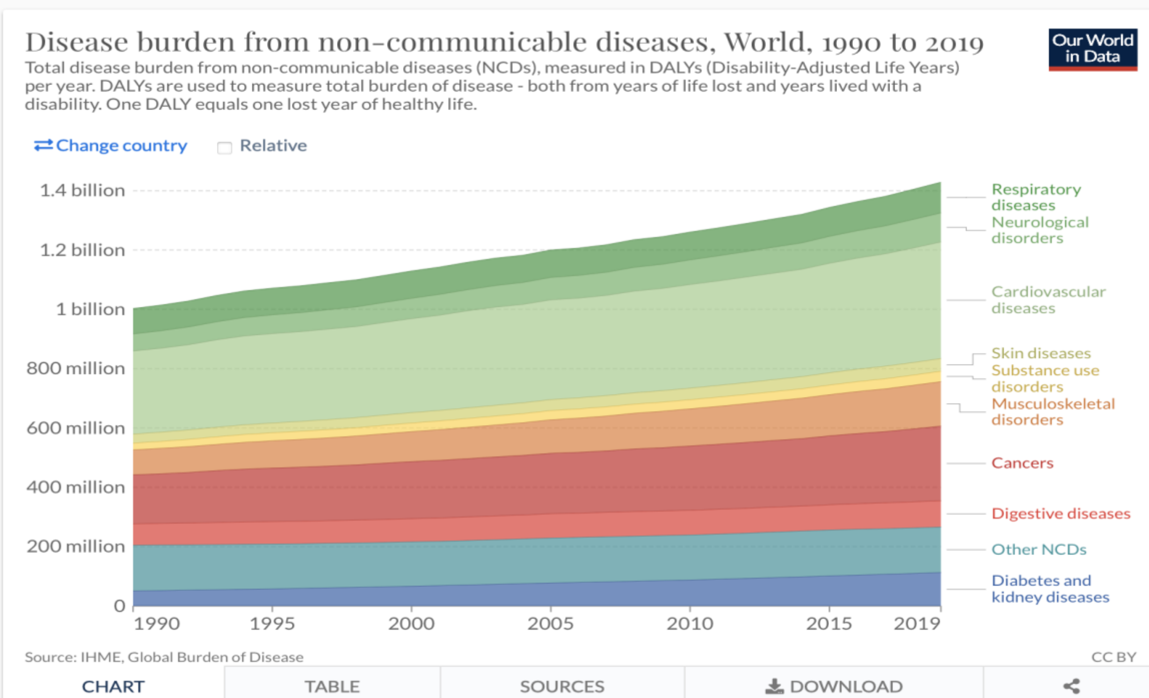

Burden of disease by cause, Albania, 2019
Our World in Data
Total disease burden, measured in Disability-Adjusted Life Years (DALYs) by sub-category of disease or injury.
DALYs measure the total burden of disease – both from years of life lost due to premature death and years lived with a disability. One DALY equals one lost year of healthy life.
Change country
Cardiovascular diseases 221,259.93
Cancers 111,719.01
Musculoskeletal disorders 54,290.6
Unintentional injuries 48,445.83
Other NCDs 46,656.41
Neurological disorders 46,451.3
Mental disorders 40,906.84
Diabetes and kidney diseases 25,818.06
Transport injuries 21,568.08
Respiratory diseases 19,951.39
Digestive diseases 19,444.51
Neonatal disorders 19,261.89
Respiratory infections and TB 15,560.62
Skin diseases 11,758.89
Substance use disorders 8,762.58
Self-harm 7,244.07
Nutritional deficiencies 6,697.16
Enteric infections 5,220.17
Interpersonal violence 4,312.03
Other infectious diseases 3,612.43
Natural disasters 2,292.43
Conflict and terrorism 700.76
HIV/AIDS and STIs 560.3
Malaria & neglected tropical diseases 483.59
Maternal disorders 401.26
0
50,000
100,000
150,000
200,000
Source: IHME, Global Burden of Disease
OurWorldInData.org/burden-of-disease • CC BY
Note: Non-communicable diseases are shown in blue; communicable, maternal, neonatal and nutritional diseases in red; injuries in grey.
1990
2019
CHART
TABLE
SOURCES
DOWNLOAD


Burden of disease, by age group, World, 1990 to 2019
Our World in Data
Disability-Adjusted Life Year (DALYs) from all causes per 100,000 individuals, by age group.
DALYs measure the total burden of disease – both from years of life lost due to premature death and years lived with a disability. One DALY equals one lost year of healthy life.
Change country
Relative change
160,000
140,000
120,000
100,000
80,000
60,000
40,000
20,000
0
2011
70+ years old 110,444
Under-5s 98,900
50-69 year olds 52,361
Age-standardized 37,550
All ages 36,114
15-49 year olds 21,458
5-14 year olds 10,470
50-69 year olds
Age-standardized
All ages
15-49 year olds
5-14 year olds
1990
1995
2000
2005
2010
2015
2019
Source: IHME, Global Burden of Disease
OurWorldInData.org/burden-of-disease • CC BY
1990
2019
CHART
TABLE
SOURCES
DOWNLOAD

The disease burden from non-communicable diseases



The burden from non-communicable diseases by sub-category



Project Schedule:

Time	**Activity**
16/04/2022	Gathering, Cleaning and Shaping Data Completed
27/04/2022	Creation of All Visualizations and Including various details to the visualizations
02/05/2022	Compile all graphs in an initial presentation draft for review
08/05/2022	Make all changes identified from draft review.

Data Collection:
Data is obtained from Ourworldindata.org. The project is modeled after the article below, Max Roser and Hannah Ritchie (2021) - "Burden of Disease". Published online at OurWorldInData.org. Retrieved from: https://ourworldindata.org/burden-of-disease [Online Resource]. All data is obtained from the various sections and will be shaped correctly to create the required visualizations.

Data Links:

1. Data set contains countries and DALY results for 1990-2019 as well as relative and absolute changes over time. Data set is well structured. There might be need of removing certain regions that are not graphable.
https://ourworldindata.org/grapher/dalys-rate-from-all-causes?tab=table
2. Data set contains the various disease and their contribution to DALY for each country and over the period 1990 -2019.
https://ourworldindata.org/grapher/burden-of-disease-by-cause?tab=table

3. Data set compares DALY amongst various ages and across the period 1990 - 2019.There will be a need to separate the data for various countries to make visualization easier.
https://ourworldindata.org/grapher/daly-rates-from-all-causes-by-age?tab=table&country=~OWID_WRL

4. Data set details DALY rates from communicable, neonatal , maternal and nutritional disease 1990 – 2019. The data will be broken down for various headings.
https://ourworldindata.org/grapher/burden-of-disease-rates-from-communicable-neonatal-maternal-nutritional-diseases?tab=table

5. Data set details DALY rates from non-communicable diseases for 1990 – 2019. The data will be broken down for various diseases.
https://ourworldindata.org/grapher/burden-of-disease-rates-from-ncds?tab=table

6. Data set contains the DALY rates for injuries from 1990 -2019 for various countries.
https://ourworldindata.org/grapher/burden-of-disease-rates-from-injuries?tab=table

Reference:

Max Roser and Hannah Ritchie (2021) - "Burden of Disease". Published online at OurWorldInData.org. Retrieved from: https://ourworldindata.org/burden-of-disease [Online Resource]